



Parkinson's disease classification using gait characteristics and wavelet-based feature extraction

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ARTICLE INFO

Keywords:

Parkinson's disease
Gait
Fuzzy neural networks
Wavelet transforms
Feature extraction

ABSTRACT

This paper proposes a method to classify idiopathic PD patients and healthy controls using both the gait characteristics of idiopathic PD patients and wavelet-based feature extraction. Using the characteristics of idiopathic PD patients who shuffle their feet while they are walking, we implemented the following three preprocessing methods: (i) we used the difference between two signals that each represented the sum of eight sensor outputs from one foot; (ii) we used the difference between the maximum and minimum records among the vertical ground reaction force outputs from the eight sensors under the left foot; and (iii) we used method (i) again, but on the signals each obtained from one foot by method (ii). After thus conducting the three preprocessing tasks, we created approximation coefficients and detail coefficients using wavelet transforms (WTs). Then, we extracted 40 features from these coefficients by using statistical approaches, including frequency distributions and their variabilities. Using the 40 features as inputs to the neural network with weighted fuzzy membership functions (NEWFM), we compared the performances of the three abovementioned methods. When idiopathic PD patients and healthy controls were classified using the NEWFM, the accuracy, specificity, and sensitivity of the results were, respectively, as follows: 74.32%, 81.63%, and 73.77% by method (i); 75.18%, 74.67%, and 75.24% by method (ii); or 77.33%, 65.48%, and 81.10% by method (iii).

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1. Introduction

Parkinson's disease is a representative degenerative disease that occurs due to a deficiency of nerve cells, called dopamine neurons, distributed in the substantia nigra of the brain (Koller & et al., 1989). The four major characteristic symptoms that indicate Parkinson's disease are resting tremor, rigidity, postural instability, and bradykinesia (Koller et al., 1989; Morris, 2000). In particular, as clinical symptoms of bradykinesia, diverse motor disturbances appear, such as remarkable decreases in moving speeds, reductions in arm swinging while walking, and shuffling gaits (Echeverria, Rodriguez, Velasco, & Alvarez-Ramirez, 2010; Henmia et al., 2006; Lewek, Poole, Johnson, Halawa, & Huang, 2010; Patterson, Gage, Brooks, Black, & McIlroy, 2010; Yang, Lee, Cheng, Lin, & Wang, 2008).

To classify Parkinson's disease through bradykinesia, the Unified Parkinson's Disease Rating Scale (UPDRS) (Martinez-Martin et al., 1994; Richards, Marder, Cote, & Mayeux, 1994), finger motion analysis (Jobbágy, Harcos, Karoly, & Fazekas, 2005; Kim et al., 2010), and foot pressure analysis (Kim & Eom, 2006; Lee et al., 2007) are used. However, the UPDRS, which is a representative clinical test scale,

not only relies on the experience of skilled clinicians, but also produces subjective judgments (Fahn et al., 1987; Machado et al., 2006). In addition, the UPDRS has another disadvantage in that a long time is required to measure all the items required (Post, Merkus, de Bie, de Haan, & Speelman, 2005). The methods of finger motion analysis or foot pressure analysis begin with data collection but mainly use statistical approaches offline. These statistical approaches involve a weakness in that judgments are not made objectively by systems on the basis of certain rules, but rather are made subjectively by persons on the basis of statistical results. Thus, the resultant judgments may vary with the opinions of the judges. Therefore, the foregoing methods to classify Parkinson's disease cannot provide general rules that enable interpretation of inputs but merely produce subjective judgments through statistical approaches.

In this study, we extracted wavelet-based features using the gait characteristics of idiopathic PD patients. Then, we used the features thus extracted as inputs of the neural network with weighted fuzzy membership functions (NEWFM) (Lee & Lim, 2011; Lim, 2009; Lim, Wang, Kim, & Gupta, 2006) to classify both idiopathic PD patients and healthy controls. Thereafter, we enabled interpretation of the inputs by using the bounded sum of the weighted fuzzy membership functions (BSWFM) as provided by the NEWFM. In the first stage, to extract inputs for the NEWFM, we used the vertical ground reaction force on each of the 16

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sensors located under the feet of each idiopathic PD patient and healthy control. In the second stage, using the gait characteristics of idiopathic PD patients who shuffle their feet, we implemented the following three preprocessing methods: (i) we used the difference between two signals that each represented the sum of eight sensor outputs from one foot; (ii) we used the difference between the maximum and minimum records among the vertical ground reaction force outputs from the eight sensors under the left foot; and (iii) we used method (i) again, but on the signals each obtained from one foot by method (ii). In the third stage, we used wavelet transforms (WT) to extract approximation coefficients and detail

coefficients from the preprocessed data of the second stage. Then, we extracted 40 features from the approximation coefficients and detail coefficients by using statistical approaches, including frequency distributions and their variabilities. In the final stage, we used the 40 features extracted in the third stage as inputs to the NEWFM to compare the performances of the three methods.

When idiopathic PD patients and healthy controls were classified using the NEWFM, the accuracy, specificity, and sensitivity of the results were, respectively, as follows: 74.32%, 81.63%, and 73.77% by method (i); 75.18%, 74.67%, and 75.24% by method (ii); or 77.33%, 65.48%, and 81.10% by method (iii). Based on these

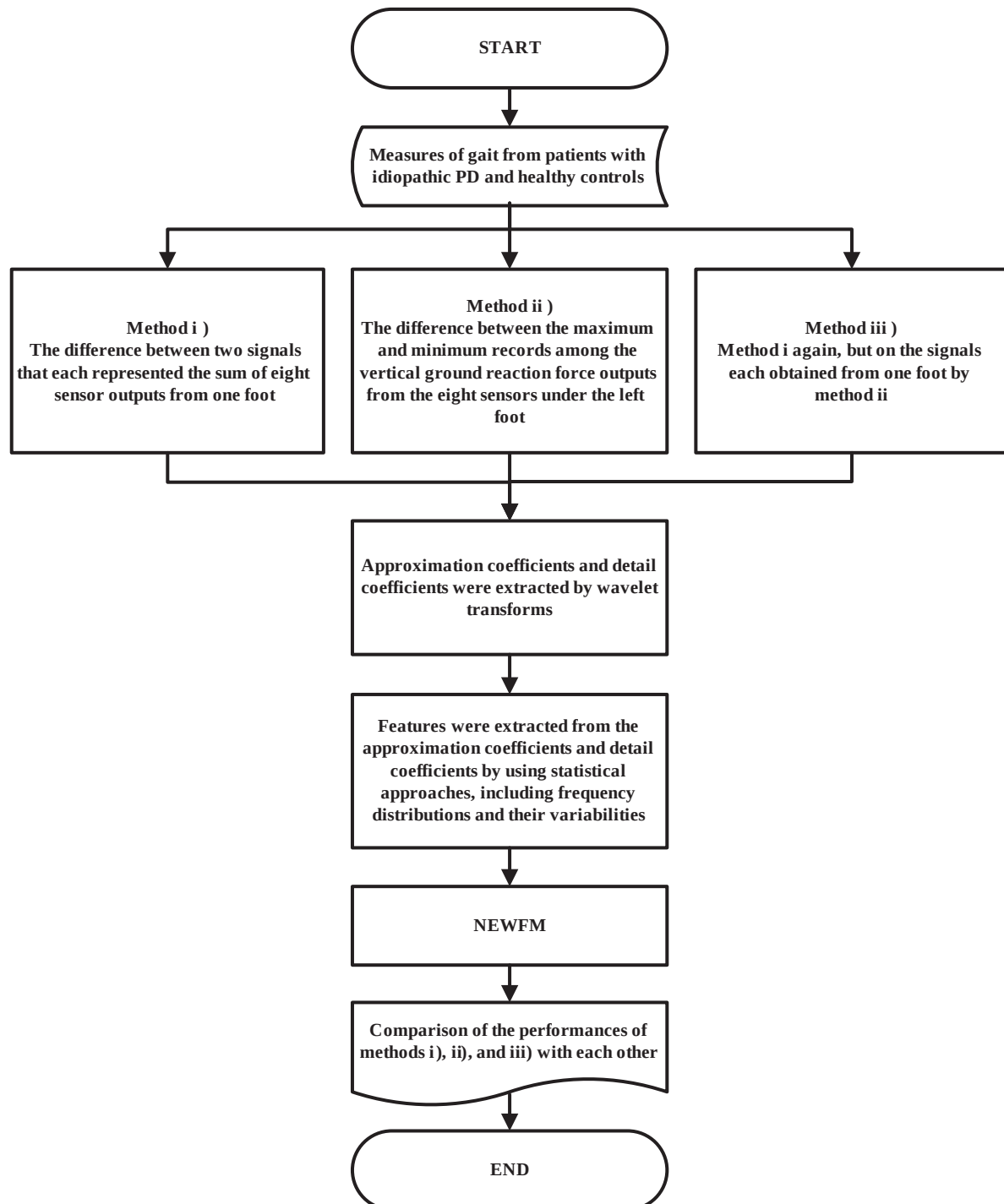


Fig. 1. Diagram of models proposed for classifying idiopathic PD patients and healthy controls.

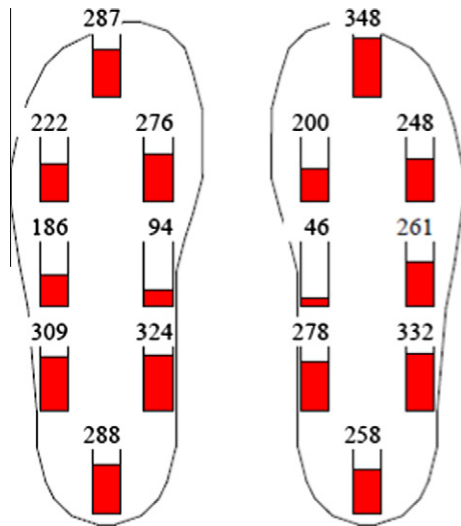


Fig. 2. Examples of vertical ground reaction force records for each of eight sensors located under each foot.

Table 1
Description of measured data.

Input column	Variable	Input description
1	t_n	Time (in seconds)
2–9	$l_{n1}, l_{n2}, \dots, l_{n8}$	Vertical ground reaction force records for each of the eight sensors located under the left foot as shown in Fig. 2
10–17	$r_{n1}, r_{n2}, \dots, r_{n8}$	Vertical ground reaction force records for each of the eight sensors located under the right foot as shown in Fig. 2
18	fl_n	Total force under the left foot
19	fr_n	Total force under the right foot

results, method (iii) that combined method (i) and method (ii) showed the highest accuracy. In this study, the 40 features were interpretably formed into weighted fuzzy membership functions preserving the disjunctive fuzzy information and characteristics (Lee & Lim, 2011; Lim, 2009; Lim et al., 2006).

2. Overview of models proposed for classifying idiopathic PD patients and healthy controls

As shown in Fig. 1, a diagram of the models proposed for classifying idiopathic PD patients and healthy controls, we implemented the three methods using the gait characteristics of idiopathic PD patients from vertical ground reaction force records collected from sensors.

Once wavelet transforms (WT) had been taken, we extracted features of the approximation coefficients and detail coefficients using the frequency distributions and their variabilities. The features thus extracted were used as inputs to the NEWFM to compare the abilities of the three methods to classify idiopathic PD patients and healthy controls.

Table 3
Statistics of Wilcoxon rank-sum test.

p -value (two-sided)	0.000
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2.1. Experimental data

In this study, a database provided by PhysioBank (<http://www.physionet.org/physiobank/database/gaitpdb/>) was used. This database contains measures of gait for 93 idiopathic PD patients (mean age 66.3 years, 63% men) and 73 healthy controls (mean age 66.3 years, 55% men). The database includes the vertical ground reaction force records of subjects as they walked at their self-selected pace for approximately 2 min on level ground. As shown in Fig. 2, underneath each foot were eight sensors that measured forces (in Newtons) as functions of time. The individual outputs of these 16 sensors (input columns 2–17 in Table 1) were digitized and recorded at 100 samples per second, but the records also include two signals (input columns 18 and 19 in Table 1) that each represent the sum of the eight sensor outputs from one foot.

Table 1 describes the vertical ground reaction force records, and their sums, measured using the eight sensor outputs from each foot every 0.01 s. This database stores 92 files collected from 73 healthy controls and 214 files collected from 93 idiopathic PD patients.

2.2. Preprocessing

2.2.1. Preprocessing using gait characteristics

In this study, the 19 inputs described in Table 1 were used to implement the three methods described in Table 2 by using the gait characteristics of idiopathic PD patients as illustrated in Fig. 1. From each database file, 2048 data were extracted in order and later data were discarded. Since the data were stored at a rate of 100 samples per second, 2048 data corresponded to approximately 20 s.

Table 3 shows the result of a Wilcoxon rank-sum test (Lee, Lim, & Shin, 2010). The p -value was smaller than 0.05 for a significance level of 5%, and therefore, the null hypothesis that the vertical ground reaction force records of idiopathic PD patients and healthy controls would be the same as each other was dismissed. Therefore, classifying idiopathic PD patients and healthy controls using the vertical ground reaction force records can be regarded as meaningful.

2.2.2. Feature extraction using wavelet transform

By analyzing the frequency characteristics at certain local spatial points, the wavelet transform (WT) complements the weakness of the Fourier analysis, which gives information only about global frequency characteristics. The continuous wavelet transform (CWT) of a continuous time signal $x(t)$ is defined as

$$T(a, b) = \frac{1}{\sqrt{a}} \int_{-\infty}^{\infty} \psi\left(\frac{t-b}{a}\right) x(t) dt \quad (1)$$

where $\psi((t-b)/a)$ is the analyzing wavelet function. The transform coefficients $T(a, b)$ are found for both specific locations on the signal

Table 2
Preprocessing formulas using the gait characteristics of idiopathic PD patients.

Method	Formula 1	Formula 2
i	$(fl_1 - fr_1), (fl_2 - fr_2), \dots, (fl_n - fr_n)$	$f_l(n) = \max(l_{n1}, l_{n2}, \dots, l_{n8}) - \min(l_{n1}, l_{n2}, \dots, l_{n8})$
ii	$f_l(1), f_l(2), \dots, f_l(n)$	$f_r(n) = \max(r_{n1}, r_{n2}, \dots, r_{n8}) - \min(r_{n1}, r_{n2}, \dots, r_{n8})$
iii	$f_l(1) - f_r(1), f_l(2) - f_r(2), \dots, f_l(n) - f_r(n)$	

Notes: $n = 2048$.

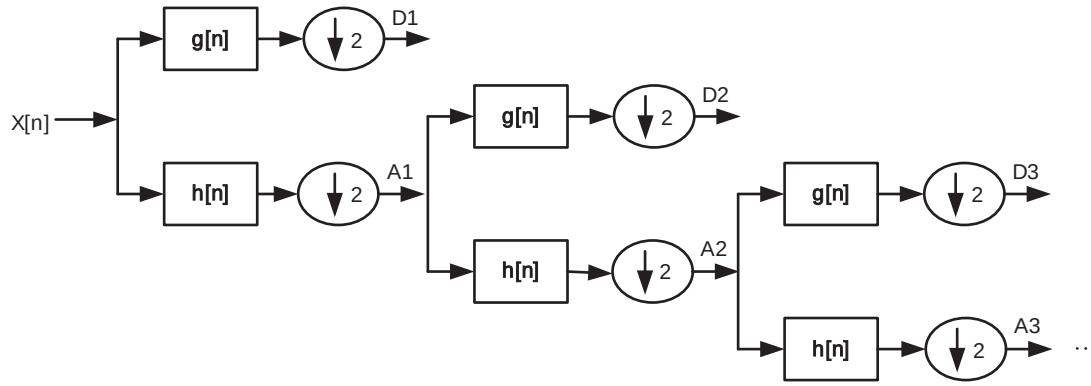


Fig. 3. Sub-band decomposition of discrete wavelet transforms; $g[n]$ is the high-pass filter, $h[n]$ is the low-pass filter.

Table 4

Description of feature extraction.

No	Description of feature extraction
1	Mean of the absolute values of the coefficients in each sub-band
2	Median of the coefficients in each sub-band
3	Average power of the wavelet coefficients in each sub-band
4	Standard deviation of the coefficients in each sub-band
5	Ratio of the absolute mean values of adjacent sub-bands

Table 5

Numbers of training and testing sets.

Class	Training sets	Testing sets	Total sets
Idiopathic PD patients	487	487	974
Healthy controls	210	210	420
Total sets	697	697	1394

($t = b$) and for specific wavelet periods (functions of the scale of a). The CWT becomes the dyadic wavelet transform (DWT) if a is discretized along the dyadic sequence 2^i where $i = 1, 2, \dots$. The DWT can be defined as follows (Mallat, 1991):

$$S_{2^i}x(n) = \sum_{k \in \mathbb{Z}} h_k S_{2^{i-1}}x(n - 2^{i-1}k) \quad (2)$$

$$W_{2^i}x(n) = \sum_{k \in \mathbb{Z}} g_k S_{2^{i-1}}x(n - 2^{i-1}k) \quad (3)$$

where S_{2^i} is a smoothing operator; W_{2^i} is the digital signal $x(n)$; $i \in \mathbb{Z}$ (the integral set); and h_k and g_k (see Fig. 3) are coefficients of the corresponding low-pass and high-pass filters. A filtered signal at level i is downsampled, reducing the length of the signal at level $i - 1$ by a factor of two, and generating the approximation coefficients (a_i) and detail coefficients (d_i) at level i .

In this study, the Haar WTs, which are at scale level 5, were taken with the values extracted using methods (i, ii, and iii) (Lee et al., 2010). From the detail coefficients and approximation coefficients, which are then wavelet coefficients at levels 2–5, we extracted 40 features using the statistical approaches described in Table 4 and we subsequently used those as inputs. As explained in Table 4, features 1, 2, and 3 represent the frequency distributions of the values extracted using the three methods, while features 4 and 5 represent the amounts of variability in those frequency distributions (Kandaswamy, Sathish Kumar, Ramanathan, Jayaraman, & Malmurugan, 2004; Lee & Lim, 2009a; Lee et al., 2010).

We composed experimental groups each consisting of the 40 inputs extracted by the statistical approaches described in Table 4. Specifically, as shown in Table 5, we composed 420 experimental

groups of healthy controls and 974 experimental groups of idiopathic PD patients. However, we divided the experimental groups into a first half and latter half as training sets and testing sets, respectively, based on the arrangement of the files in storage.

2.3. Neural network with weighted fuzzy membership functions (NEWFM)

In this study, a neural network with weighted fuzzy membership functions (NEWFM) was used for classifying idiopathic PD patients and healthy controls. The NEWFM is a supervised classification neuro-fuzzy system that uses the bounded sum of weighted fuzzy membership functions (BSWFM) (Lee & Lim, 2011; Lim, 2009; Lim et al., 2006). The structure of the NEWFM, as shown in Fig. 4, is composed of three layers; namely, the input, hyperbox, and class layers. Using the feature extraction method explained in Table 4, 40 features were extracted from the approximation coefficients and detail coefficients, and these 40 features were in turn used as inputs to the NEWFM, as shown in Fig. 4.

The NEWFM has the advantage that it provides a non-overlapping area distribution measurement method to enable selection of the minimum features (Lee & Lim, 2009b, 2011; Lim, 2009). Feature selection was implemented for better performance, such as higher accuracy and lower operating cost, by using minimum features.

As examples of feature selections, Figs. 5 and 6 describe good candidate features and bad candidate features, respectively. Since the two BSWFMs in Fig. 5 have a larger overlapping area than the BSWFMs in Fig. 6, they are considered somewhat better features. Therefore, the larger the non-overlapping area, the more a feature selection is implied. In this study, however, although we conducted feature selections using a non-overlapping area distribution measurement method provided by the NEWFM (Lee & Lim, 2009b, 2011; Lim, 2009), the best performance was exhibited when the features selected were the 40 initial ones.

Figs. 5 and 6 show the bounded sum of the weighted fuzzy membership functions (BSWFM) for two of the 40 features. Through this, differences between healthy controls and idiopathic PD patients can be tangibly seen among the 40 features and related characteristics can be analyzed.

3. Experimental results

In this study, the performances of the three methods described in Table 2 were compared with each other and analyzed. The performance comparisons between methods (i, ii, and iii) are shown in Tables 6–8 using four performance classifications. TP (true positive) refers to cases where idiopathic PD patients were classified

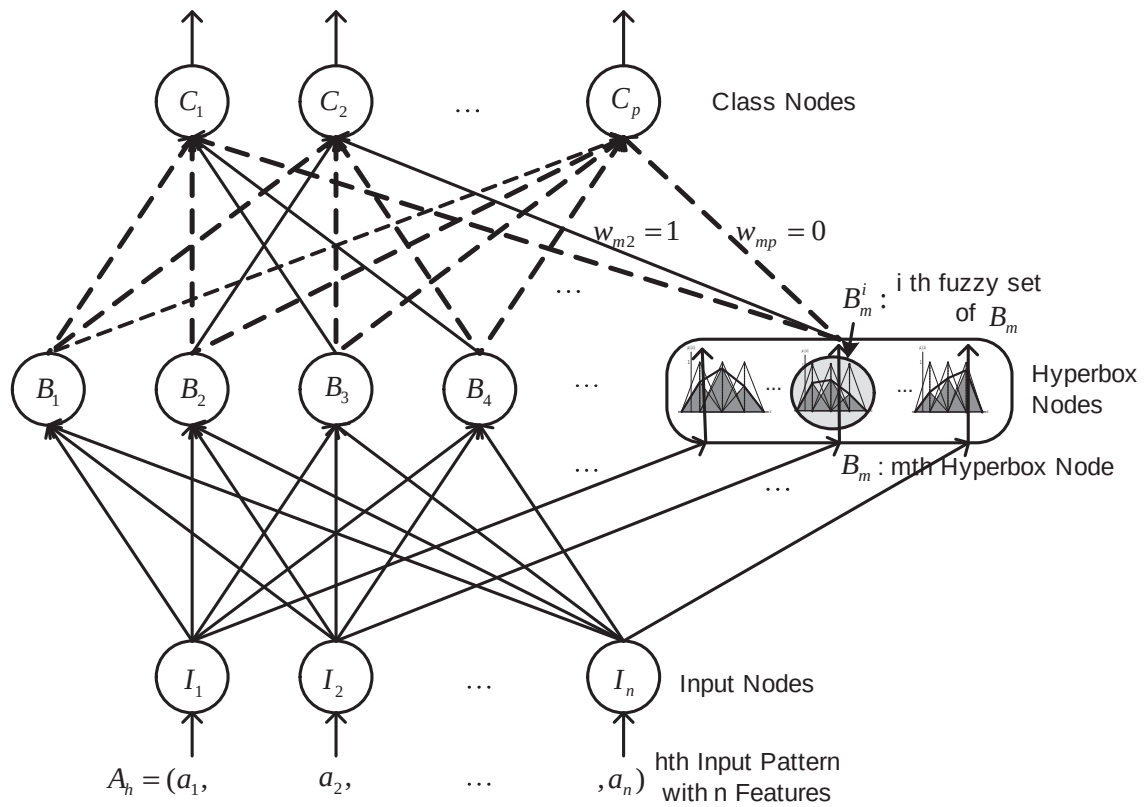


Fig. 4. Structure of NEWFM.

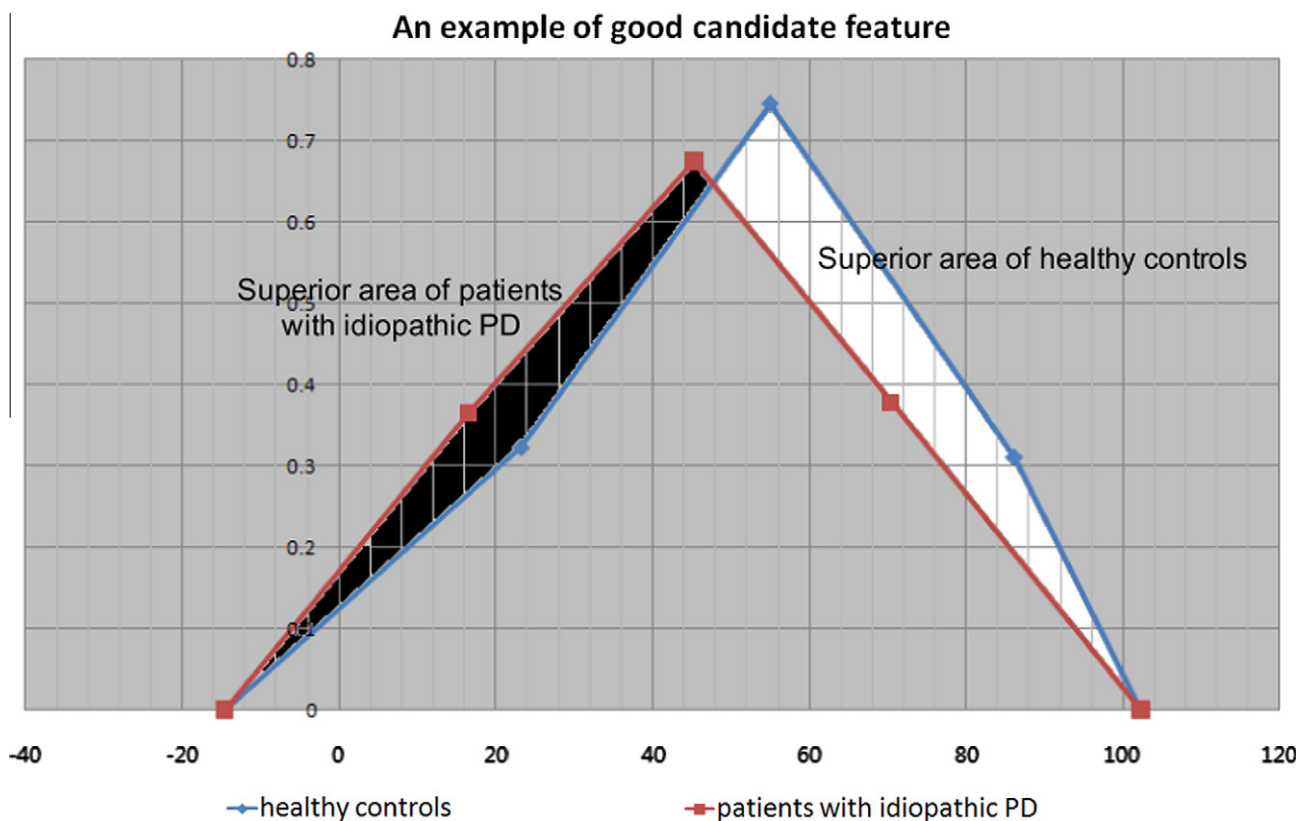


Fig. 5. An example of a good candidate for feature selection (standard deviation of the coefficients at level 2).

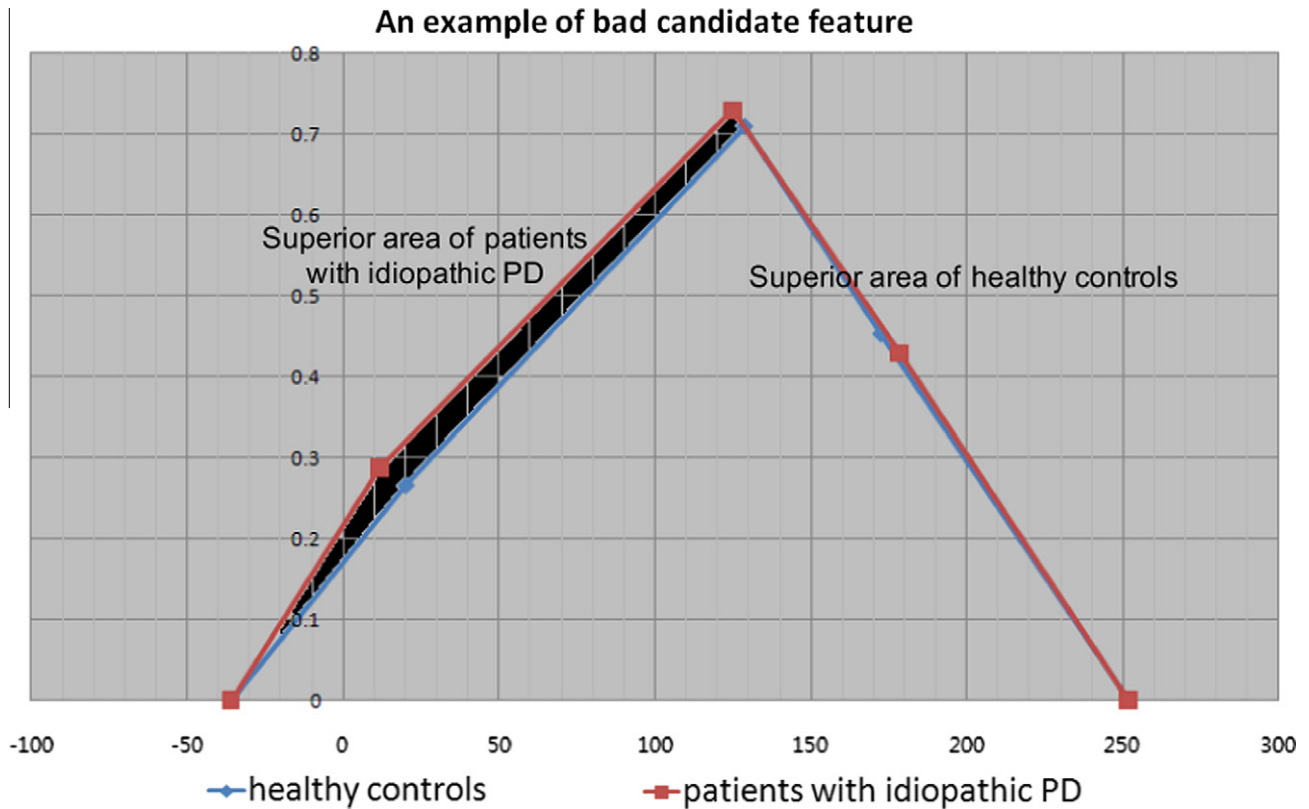


Fig. 6. An example of a bad candidate for feature selection (standard deviation of the coefficients at level 3).

Table 6

Confusion matrix of classification results (method (i)).

Idiopathic PD patients	TP 478	FN 170
Healthy controls	FP 9	TN 40

Table 7

Confusion matrix of classification results (method (ii)).

Idiopathic PD patients	TP 468	FN 154
Healthy controls	FP 19	TN 56

Table 8

Confusion matrix of classification results (method (iii)).

Idiopathic PD patients	TP 429	FN 100
Healthy controls	FP 58	TN 110

as idiopathic PD patients, and TN (true negative) refers to cases where healthy controls were classified as healthy controls. FP (false positive) refers to cases where idiopathic PD patients were classified as healthy controls, and FN (false negative) refers to cases where healthy controls were classified as idiopathic PD patients. The sensitivities, specificities, and accuracies that are listed in Table 9 were obtained as follows:

Table 9

Comparison of performance results for methods (i, ii, and iii).

	Accuracy (%)	Specificity (%)	Sensitivity (%)
Method (i)	74.32	81.63	73.77
Method (ii)	75.18	74.67	75.24
Method (iii)	77.33	65.48	81.10

$$\text{Sensitivity} = \frac{TP}{TP + FN} \times 100$$

$$\text{Specificity} = \frac{TN}{TN + FP} \times 100 \quad (4)$$

$$\text{Accuracy} = \frac{TP + TN}{TP + FN + TN + FP} \times 100$$

4. Concluding remarks

This paper proposes a method to classify idiopathic PD patients and healthy controls using the gait characteristics of idiopathic PD patients and wavelet-based feature extraction. Using the characteristics of idiopathic PD patients who shuffle their feet while they are walking, we implemented a preprocessing method. In the preprocessing task, we obtained the difference between the maximum and the minimum of the eight vertical ground reaction force records from the sensors located under the left foot, and similarly the right foot, before we then obtained the difference between the two differences. After thus conducting the preprocessing task, we created approximation coefficients and detail coefficients using wavelet transforms (WTs), and we then extracted 40 features from these coefficients by using statistical approaches, including frequency distributions and their variabilities. Using the 40 features as inputs of the NEWFM, we assessed the performance of the

method. When idiopathic PD patients and healthy controls were classified using the 40 features as inputs, the accuracy, specificity, and sensitivity of the results were 77.33%, 65.48%, and 81.10% respectively. Through the method proposed in this paper, a system that can classify idiopathic PD patients and healthy controls in real time can be implemented. The NEWFM obtains the bounded sum of weighted fuzzy membership functions (BSWFM) for the 40 features to reveal the fuzzy membership functions for the 40 features. These fuzzy membership functions are used to classify idiopathic PD patients and healthy controls.

Acknowledgements

This work was supported by the Gachon University Research Fund of 2011 (KWU-2011-R346).

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